

Manthan Bhala

Machine Learning Engineer & Software Developer

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SUMMARY

Machine learning engineer and software developer building production systems for search, ranking, and recommendations. I build practical ML systems that improve e-commerce discovery, with a focus on retrieval, ranking, and recommendation pipelines — turning model work into reliable software and measurable product impact through thoughtful experimentation, evaluation, and production-ready implementation.

EXPERIENCE

Senior ML Engineer — Makro PRO

Dec 2023 – Present

Gurugram, Haryana, India

- Build machine learning systems for e-commerce search, ranking, and recommendation workflows.
- Work on model distillation, feature design, evaluation, and production rollout to improve customer discovery and conversion.

KEY ACHIEVEMENTS

Personalized ranking via LLM distillation — LLM4Rank + PEFT → LightGBM delivering a **43.9% conversion lift** and a **52.6% add-to-basket lift**.

E-commerce recommendation system delivering **28% conversion** and more than **70,000 weekly conversions**.

SKILLS

ML Engineering

Python · PyTorch · TensorFlow · Hugging Face

Applied AI

Information Retrieval · Learning to Rank · Recommendation Systems · LLMs · PEFT

SELECTED INTELLECTUAL PROPERTY

Patent 2501007631 — A Hybrid Lexical-Semantic Retrieval Process for E-Commerce Search

Patent 2501004476 — A Hybrid LLM Process for Multi-Objective Reranking in E-Commerce

Patent 2501004419 — A Multilingual Phonetic BK-Tree Based Spelling Correction Process in E-Commerce